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Peaks or thresholds? Non-Gaussian primordial black hole abundances for broad spectra and the pulsar-timing-array tension

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Abstract. Primordial black holes (PBHs) from a peak in the curvature spectrum are counted either with threshold statistics or with peaks theory. With local non-Gaussianity (NG), peaks theory had been applied only to monochromatic spectra or with the peak condition imposed on a Gaussian field. We construct it on the compaction function for broad spectra, as the exact counterpart of threshold statistics. Maxima of the non-Gaussian compaction $C_\ell = F'(\zeta_r)C_\mathrm{g}$ are maxima of the Gaussian field $C_\mathrm{g} + \lambda\zeta_r$, times a rank-one Hessian factor that we compute to first order; a Monte Carlo and counts on 256^3 lattices confirm it, and show that the Gaussian-peak approximation miscounts peaks by up to 18%. The amplitude ratio $R = A_*^\mathrm{peaks}/A_*^\mathrm{threshold}$ is 0.74–0.82 for $|f_\mathrm{NL}| \leq 2$ and lognormal widths 0.1–2, and NG cancels in R/R_Gauss to about 4% (16% for a strong ultra-slow-roll tail), so one function of the width converts threshold-statistics results to peaks theory. For the NANOGrav 15-year data, peaks theory strengthens the exclusion of a scalar-induced gravitational-wave explanation by 0.2–0.8σ for every NG template, or by 0.04–0.39σ with a background from supermassive black hole binaries, which caps the exclusion at 2.8σ. Primordial NG in the induced gravitational waves, with all seven terms, lowers the significance by up to 0.26σ and leaves this shift unchanged. Three rarely stated analysis choices matter as much as the statistics: the frequency range, when the transfer function acts in the collapse threshold, and the normalisation of f_PBH, which the expression in common use overestimates by a factor of 3–13.

Keywords: primordial black holes, non-gaussianity, gravitational waves / theory, physics of the early universe

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1 Introduction

Pulsar timing arrays (PTAs) have found evidence for a nanohertz gravitational-wave (GW) background [1–4]. One cosmological interpretation is the GWs induced at second order by a peak in the primordial curvature spectrum [5–11]. A peak large enough to produce the signal also forms primordial black holes (PBHs) when the perturbations re-enter the horizon, and requiring that PBHs are not overproduced disfavours the scenario at the 2–3σ level [9, 12, 13]. How strongly depends on the non-Gaussianity (NG) of the curvature perturbation [12, 14, 15] and on how the PBH abundance is computed [13].

Two prescriptions are in use for the abundance (see ref. [16] for a review). Threshold statistics counts the volume fraction in which the compaction function [17, 18] exceeds a threshold; it includes the non-linear relation between curvature and density [19, 20] and any local NG non-perturbatively [21, 22]. Peaks theory counts maxima instead [23, 24]. It has

been developed for Gaussian curvature perturbations with spectra of finite width [25–29] and, with local NG, either for monochromatic spectra [30–32] or with the peak condition imposed on the Gaussian component [33–35]. For Gaussian curvature perturbations the two prescriptions have been compared in refs. [13, 22, 28, 36]. Several recent papers state that peaks theory with local NG for broad spectra is an open problem [13, 22, 27, 29, 37, 38], and Franciolini et al. [12] note that peaks theory "provides slightly smaller amplitudes", so that their threshold-statistics bounds are conservative.

This paper fills that gap and quantifies their remark. In section 2 we build peaks theory on the compaction function for broad spectra with local NG (quadratic, logarithmic or ultra-slow-roll, and curvaton templates), as the exact counterpart of threshold statistics, so that the two differ only by the peak condition. The main technical step is to count maxima of the non-Gaussian compaction itself, a local function of two correlated Gaussian fields, rather than of its Gaussian component. Section 3 sets out the conventions for the abundance. Section 4 gives the amplitude ratio $R(\Delta, \mathrm{NG})$ between the statistics, and section 5 propagates it to the significance of the PBH-overproduction tension for NANOGrav 15, with every analysis choice stated and varied. There we also add the primordial-NG corrections to the GW spectrum [39–41], which ref. [12] neglects, and an astrophysical background. Section 6 discusses the results and their limitations, and section 7 concludes.

Two conventions turn out to matter, and the second is widespread in the literature. The collapse threshold of ref. [12] follows the superhorizon smoothing of Musco et al., in effect without a transfer function, although their formula is written with the horizon-crossing transfer function; read that way, it changes the amplitude by 18–62% (section 3 and appendix C). The normalisation of f_{PBH} in common use is too large by a factor of 3–13 (appendix B).

Our construction differs from the closest existing work as follows. Yoo, Gong and Yokoyama [34] count peaks of ζ_{g} with typical profiles and quadratic NG, assuming that every peak of ζ_{g} is a peak of ζ, which holds for $\frac{3}{5}f_{\mathrm{NL}} \gtrsim -\frac{1}{4}$. Riccardi et al. [35] treat quadratic NG for finite-width spectra with peaks of the curvature perturbation and the linear relation between density and curvature, not the compaction function. Escrivà [42] counts peaks of ζ_{g} for the logarithmic template and adds a dispersion of the profile shape, which for negative NG and broad spectra raises the abundance by orders of magnitude. Germani et al. [43] give the statistics of peaks of general local-NG fields and apply it to their sphericity. None of these compares with threshold statistics in the same conventions. We count maxima of the non-Gaussian compaction itself, with the threshold, window, transfer function and cut of threshold statistics, so that R isolates the peak condition.

2 Two statistics for the non-Gaussian compaction

2.1 Local NG and the compaction function

The curvature perturbation is $\zeta = F(\zeta_{\mathrm{g}})$ with ζ_{g} Gaussian and power spectrum $\mathcal{P}_\zeta$. We use three templates: the quadratic $F = \zeta_{\mathrm{g}} + \frac{3}{5}f_{\mathrm{NL}}\zeta_{\mathrm{g}}^2$; the ultra-slow-roll (USR) logarithm $F = -\mu_* \ln(1 - \zeta_{\mathrm{g}}/\mu_*)$ [33], which is the $F = -\frac{2}{\beta}\ln(1 - \frac{\beta}{2}\zeta_{\mathrm{g}})$ of ref. [12] with $\mu_* = 2/\beta$ and has $f_{\mathrm{NL}} = 5/(6\mu_*)$ at second order; and the sudden-decay curvaton, $\zeta = \ln X(r_{\mathrm{dec}}, \zeta_{\mathrm{g}})$ [21, 44]. For the curvaton, X depends on ζ_{g} only through $(1 + 3\zeta_{\mathrm{g}}/2r_{\mathrm{dec}})^2 = (\sigma/\bar{\sigma})^2$, with σ the curvaton field, so troughs of ζ_{g} below $-2r_{\mathrm{dec}}/3$, where σ changes sign, are also peaks of ζ. For $r_{\mathrm{dec}} < 1$ we integrate both branches.

We use two spectral shapes, a lognormal of width Δ and the broken power law BPL(α,β,γ) of ref. [12]:

$$\mathcal{P}_\zeta = \frac{A}{\sqrt{2\pi}\,\Delta}\exp\Big[-\frac{\ln^2(k/k_*)}{2\Delta^2}\Big], \qquad \mathcal{P}_\zeta = A\,\frac{(\alpha+\beta)^\gamma}{\big[\beta(k/k_*)^{-\alpha/\gamma}+\alpha(k/k_*)^{\beta/\gamma}\big]^\gamma}. \tag{2.1}$$

A is the integrated power of the lognormal and the peak value of the BPL. The slopes α,β and the width γ of the BPL are not to be confused with the mass fraction and the critical exponent used below. At smoothing scale r, the scale that re-enters the horizon at formation ($k = 1/r$), we define the Gaussian fields

$$C_{\rm g}(\boldsymbol{x};r) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{x}}\,\tfrac{2\Phi}{3}(kr)^2 W(kr)T(kr)\,\zeta_{\rm g}(\boldsymbol{k}), \tag{2.2}$$

$$\zeta_r(\boldsymbol{x};r) = \int \frac{\mathrm{d}^3k}{(2\pi)^3} e^{i\boldsymbol{k}\cdot\boldsymbol{x}}\,W_s(kr)T(kr)\,\zeta_{\rm g}(\boldsymbol{k}), \tag{2.3}$$

with the top-hat window $W(y) = 3(\sin y - y\cos y)/y^3$, the shell window $W_s(y) = \sin y/y$, the radiation transfer function $T(y) = W(y/\sqrt{3})$ and $\Phi = 2/3$. All moments use a hard cut $kr < 4.49$, the first zero of W, as in `pbhstat` [29]. With the transfer function the moments of the spectra used here also converge without the cut, which changes threshold-statistics amplitudes by $\leq 2.3\%$ and hardly affects the ratio between the statistics (section 4). For a spherical configuration $C_{\rm g} = -2\Phi r\zeta_{\rm g}'$ and $\zeta_r = \zeta_{\rm g}(r)$, and the linear and full compaction of $\zeta = F(\zeta_{\rm g})$ are [19–21, 30]

$$C_\ell = F'(\zeta_r)\,C_{\rm g}, \qquad C = C_\ell - \frac{C_\ell^2}{4\Phi}. \tag{2.4}$$

A PBH of mass $M = \mathcal{K}(C - C_{\rm th})^\gamma M_H$ [45–47] forms where $C > C_{\rm th}$ and $C_\ell < 2\Phi$ (type I). We use $\mathcal{K} = 3.3$ and $\gamma = 0.36$, as in refs. [21, 22]. The threshold $C_{\rm th} = \delta_c(q)$ follows the shape-dependent prescription of Musco et al. [48–50], with the transfer function in the moments.

2.2 Threshold statistics

With $P_{\rm G}(c,z)$ the bivariate Gaussian of $(C_{\rm g},\zeta_r)$, the mass fraction at one scale is [21]

$$\beta_{\rm th} = \int \mathrm{d}z \int_{C_{\ell,\rm th}}^{2\Phi} \mathrm{d}C_\ell\, \frac{P_{\rm G}(C_\ell/F'(z),z)}{|F'(z)|}\,\mathcal{K}(C - C_{\rm th})^\gamma, \tag{2.5}$$

where $C_{\ell,\rm th} = 2\Phi[1-(1-C_{\rm th}/\Phi)^{1/2}]$. Equation (2.5) includes the non-linear relation $C(C_\ell)$ and the full F.

2.3 Peaks of the Gaussian component (Gaussian-peak approximation)

The natural generalisation of earlier work is to count BBKS maxima of the Gaussian field $C_{\rm g}$, with ζ_r as a correlated scalar, and to map them through F'; where $F' < 0$, maxima of C_ℓ are minima of $C_{\rm g}$. This replaces the volume weight in eq. (2.5) by

$$w = \frac{V\,G(m,s)}{(2\pi)^{3/2}R_*^3}, \qquad G(m,s) = \int_0^\infty \mathrm{d}x\, f(x)\,\mathcal{N}(x;m,s^2), \qquad V = \tfrac{4}{3}\pi r^3, \tag{2.6}$$

where f is the BBKS function [23], m and s^2 are the conditional mean and variance of $x = -\nabla^2 C_{\rm g}/\sigma_2$ given $(C_{\rm g},\zeta_r)$, and $R_* = \sqrt{3}\sigma_1/\sigma_2$. This is exact for constant F', but not in general: $\nabla C_\ell = F'\nabla C_{\rm g} + F''C_{\rm g}\nabla\zeta_r$, so the maxima of C_ℓ are displaced from those of $C_{\rm g}$.

2.4 Peaks of the non-Gaussian compaction

The Kac–Rice density of maxima of C_ℓ at $(C_\mathrm{g}, \zeta_r) = (c, z)$ follows in three steps. First, the gradient factorises,

$$\nabla C_\ell = F'(z)\,\nabla Y_\lambda, \qquad Y_\lambda \equiv C_\mathrm{g} + \lambda\zeta_r, \qquad \lambda = \frac{cF''(z)}{F'(z)}, \tag{2.7}$$

with λ frozen at the point, so that Y_λ is a Gaussian field. Second, at a stationary point the Hessian is $\nabla\nabla C_\ell = F'[\nabla\nabla Y_\lambda + \kappa\nabla\zeta_r\nabla\zeta_r^T]$, with $\kappa = -cF'(1/F')''$; the rank-one term vanishes identically for the USR template, for which $1/F'$ is linear. Third, conditional on $\nabla Y_\lambda = 0$, the gradient $\nabla\zeta_r$ is Gaussian with variance $v = (\sigma^2_{1,cc}\sigma^2_{1,rr} - \sigma^4_{1,cr})/(3\sigma^2_{1,\lambda})$, where $\sigma^2_{n,AB} = \int \mathrm{d}\ln k\, k^{2n} K_A K_B \mathcal{P}_\zeta$ and $\sigma^2_{n,\lambda} = \sigma^2_{n,cc} + 2\lambda\sigma^2_{n,cr} + \lambda^2\sigma^2_{n,rr}$. The matrix determinant lemma then gives the peak density exactly to first order in $\tau = s\kappa v/\sigma_{2,\lambda}$ (appendix A):

$$\boxed{n(c,z) = P_\mathrm{G}(c,z)\,\frac{G(s\,m_\lambda, s_\lambda)}{(2\pi)^{3/2}R_{*,\lambda}^3}\left[1 - \tau R_2(s\,m_\lambda, s_\lambda)\right], \qquad R_{*,\lambda} = \frac{\sqrt{3}\,\sigma_{1,\lambda}}{\sigma_{2,\lambda}},} \tag{2.8}$$

with $s = \operatorname{sign} F'$, m_λ and s_λ the conditional moments of $-\nabla^2 Y_\lambda/\sigma_{2,\lambda}$, and $R_2 = 27\int f_2\mathcal{N}/\int f\mathcal{N} \to 9/m$. The peak mass fraction is eq. (2.5) with the integrand multiplied by Vn/P_G. Equation (2.8) reduces to eq. (2.6) for $\lambda = 0$ (Gaussian ζ or $F'' = 0$) and for a monochromatic spectrum, where $Y_\lambda \propto \zeta_r$, and it stays finite as $F' \to 0$. The rank-one factor is formally one power of the r.m.s. amplitude smaller, but $\kappa = -2\lambda^2/c$ for the quadratic template, and it changes the local density by up to 36% at nodes with $f_\mathrm{NL} = -2$, $\Delta = 1$.

We checked eq. (2.8) in two independent ways. A Monte Carlo of the 20 Gaussian variables with the exact Hessian reproduces it to 3×10^{-4} at 15 of 16 test nodes, and to 0.85% at the most extreme one. We also counted local maxima of C_ℓ on 256^3 Gaussian random fields (lognormal $\Delta = 1$, 24 realisations), with predictions from the lattices' own mode sums (figure 1). Equation (2.8) follows the counts to about 1% beyond the grid trend seen in the Gaussian case, while the Gaussian-peak approximation (2.6) is off by up to 18% for $f_\mathrm{NL} = -2$.

3 Abundance and its conventions

The abundance integrates the mass fraction over the horizon mass M_H at which $k = 1/r$ re-enters,

$$f_\mathrm{PBH} = \int \mathrm{d}\ln M_H\; w(M_H)\,\beta(M_H), \tag{3.1}$$

and three normalisations of the weight w are in use (appendix B):

$$w = \frac{1}{\Omega_\mathrm{DM}}\left(\frac{M_\mathrm{eq}}{M_H}\right)^{1/2} \quad \text{or} \quad w = \frac{1}{\Omega_\mathrm{DM}}\left(\frac{M_H}{M_\odot}\right)^{-1/2}\left(\frac{g_*}{106.75}\right)^{3/4}\left(\frac{g_{*s}}{106.75}\right)^{-1}\frac{1}{c_w}. \tag{3.2}$$

The first is the M_eq form of refs. [21, 29]. The second is eq. (14) of ref. [12] for $c_w = 7.9\times10^{-10}$, and the first-principles result from entropy conservation for $c_w = 1.0 \times 10^{-8}$. The first two overestimate f_PBH by factors of 3–5 and 12.7, respectively. Since β depends exponentially on $1/A$, this lowers the amplitude for $f_\mathrm{PBH} = 1$ by only $\ln(\text{factor})/|\ln\beta| \simeq 8\%$ and 12%. These factors cancel in R but not in the PTA tension.

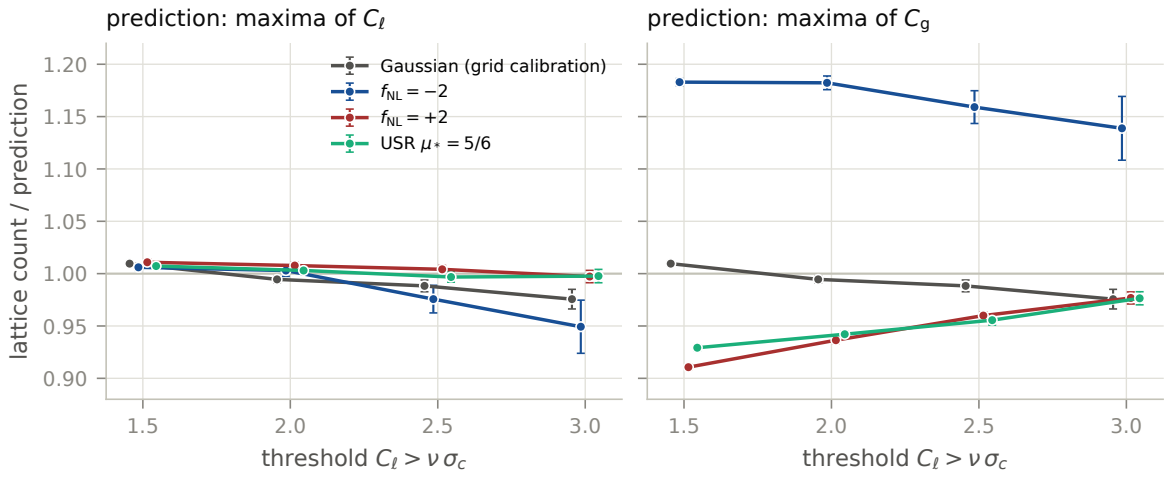


Figure 1. Test of the peak density on lattices. Maxima of the non-Gaussian compaction $C_\ell = F'(\zeta_r)C_g$ above the threshold $\nu\sigma_c$, counted on 24 Gaussian random fields of 256^3 points (lognormal spectrum, $\Delta = 1$, smoothing scale $r = 8$ cells, $\sigma_c = 0.15$, $k < 4.49/r$), divided by the prediction of eq. (2.8) (left) and of the Gaussian-peak approximation (2.6) (right), for quadratic NG with $f_{NL} = \pm 2$ and the USR template with $\mu_* = 5/6$. Both predictions use the lattices' own mode sums. Gray: Gaussian ζ, whose departure from 1 calibrates the grid.

The threshold prescription is a second, larger choice. The shape parameter q of Musco et al. [50] is computed from the spectrum smoothed with the transfer function $T(k, \eta)$, and it matters when T acts. Musco et al. apply it on superhorizon scales, $\hat{r}_m \simeq 10\, r_H$, where $T \simeq 1$ except for modes with $k\hat{r}_m \gtrsim 20$; for every spectrum used here this agrees with no transfer function to ≤ 0.0005 in δ_c. Applied at horizon crossing, as in the variances, it smooths much more: for a lognormal of width $\Delta = 1$ it gives $\delta_c = 0.537$ instead of 0.458, and for the broken power law of ref. [12] $(\alpha, \beta, \gamma = 4, 3, 1)$ it gives 0.570 instead of 0.544. Ref. [12] writes the horizon-crossing $T(k, r_m)$ in its eqs. (S4)–(S5), but its figure S3 follows the superhorizon prescription over its whole range (appendix C). The two readings move A_* by 18% (BPL) and 62% (lognormal, $\Delta = 1$) for every statistic, because the linear threshold $C_{\ell,\mathrm{th}}$ grows faster than C_{th}. They hardly affect the ratio between the statistics: R changes by $\leq 0.5\%$ for quadratic NG and by $\leq 6\%$ for the curvaton and strong USR tails. Sections 2–4 use the horizon-crossing threshold, and section 5 uses both. Table 1 collects the conventions of the two analyses.

Our implementation, `ngpbh`, reproduces the public code `pbhstat` [29] in its conventions to 0.017% in A_* for $\Delta = 0.1$–2, for both statistics. It also reproduces the broad-spectrum benchmark of Ferrante et al. (amplitude 9.09×10^{-3} against 9.2×10^{-3}) and their curvaton amplitudes to 1% [21]. The integration uses Gauss–Legendre nodes in C_ℓ that absorb the $(C - C_{\mathrm{th}})^\gamma$ cusp and an adaptive window in ζ_r, all in $\log \beta$, and it is stable to 10^{-5} when every grid is doubled.

4 The gap between the statistics, with NG

Table 2 and figure 2 give $R = A_*^{\mathrm{peaks}}/A_*^{\mathrm{threshold}}$ for $f_{\mathrm{PBH}} = 1$, with the same threshold, window, transfer function, cut, $\mathcal{K}$ and γ for both statistics. For Gaussian ζ, $R = 0.770$–0.789 in the BBKS normalisation with $V = \frac{4}{3}\pi r^3$. Over lognormal widths $0.1 \leq \Delta \leq 2$ it is fitted

	ratio R (sections 2–4)	PTA tension (section 5)
windows	top-hat W for $C_{\rm g}$, shell W_s for ζ_r	same
transfer function in the moments	$T(kr)$, radiation	same
UV cut of the moments	$kr < 4.49$	same
collapse	$\mathcal{K} = 3.3$, $\gamma = 0.36$, type I	same
threshold $C_{\rm th}$	Musco et al., T at horizon crossing	Musco et al. without T, as in ref. [12]; with T as a variant
peaks	maxima of C_ℓ, BBKS, $V = \frac{4}{3}\pi r^3$	same
$f_{\rm PBH}$ normalisation	$M_{\rm eq}$ form, eq. (3.2)	eq. (14) of ref. [12]; entropy conservation as a variant
target abundance	$f_{\rm PBH} = 1$	$f_{\rm PBH} = 1$; 10^{-3} as a variant
spectra	lognormal, $\Delta = 0.1$–2, $k_* = 10^7\,{\rm Mpc}^{-1}$; BPL$(4, \beta, 1)$, $\beta = 2$–4	BPL(4,3,1); lognormal, $\Delta = 0.5$, 1, 2
GW spectrum	—	Gaussian ζ; NG terms in section 5.4
PTA data	—	NANOGrav 15, lowest 14 bins, KDEs blurred by 0.05 dex
priors	—	uniform in $\log_{10} k_* \in [5, 10]$ and $\log_{10} A \in [-3, 0]$
astrophysical background	—	none; SMBHBs in section 5.6

Table 1. Conventions of the two analyses. The threshold, window, transfer function, cut, $\mathcal{K}$, γ and $f_{\rm PBH}$ normalisation are always the same for threshold statistics and peaks theory, so that the comparison between them is like-for-like.

to 0.002 by

$$R_{\rm Gauss} = 0.783 + 0.0066 \ln \Delta, \tag{4.1}$$

and for BPL$(4, \beta, 1)$ with $\beta = 2$–4 it is 0.775–0.778. The value of R depends on the normalisation conventions in use for peaks (the Press–Schechter factor, the peak volume and the Jacobian), across which it spans 0.75–0.89, and it is about 0.6–0.67 with the high-peak formula of ref. [13] as used in ref. [51]. These conventions cancel to better than 1% in the double ratio $R/R_{\rm Gauss}$.

The double ratio is the main result of this section. It stays within 0.958–1.036 for $|f_{\rm NL}| \leq 2$, and within 0.971–1.033 for BPL(4,3,1) (1.03 for the curvaton). Only the strongly non-perturbative USR tail $\mu_* = 5/6$ departs from it, reaching 1.05–1.16, with the largest value at $\Delta = 2$. A threshold-statistics amplitude for any of these templates can therefore be converted to peaks theory as $A_*^{\rm peaks} \simeq R_{\rm Gauss}\, A_*^{\rm threshold}$, to about 4%. For comparison, NG moves the threshold-statistics amplitude itself by factors of 0.2–2 over the same grid.

Counting the true maxima of C_ℓ rather than those of $C_{\rm g}$ matters little for the amplitude: $A_*^{(C_\ell)}/A_*^{(C_{\rm g})} = 0.983$–$1.000$, where the superscript says whose maxima are counted. The 18% miscount of peaks moves A_* by at most 1.7%, because $\delta \ln A_* \simeq \delta \ln n/|\ln \beta|$.

Two further choices could affect R. Removing the UV cut $kr < 4.49$, or moving it to $kr = 3.5$, 6 or 7.7 (the second zero of W), changes R by at most 1.2% and the double ratio by at most 0.8% (lognormal $\Delta = 0.5$–2, Gaussian and $f_{\rm NL} = \pm 2$). NG also shifts $C_{\rm th}$, by at most a few percent for $f_{\rm NL} \gtrsim -2$ [13, 52, 53]. Scaling $C_{\rm th}$ by 0.95 or 1.05 in both statistics, as in ref. [13], moves A_* by -19 to $+27$% but R and $R/R_{\rm Gauss}$ by at most 0.3% (Gaussian

template	$\Delta = 0.1$	0.3	0.5	1	2
Gaussian	0.770	0.775	0.778	0.783	0.789
$f_{\rm NL} = -2$	0.738	0.744	0.769	0.794	0.810
$f_{\rm NL} = -1$	0.748	0.751	0.760	0.784	0.803
$f_{\rm NL} = +1$	0.788	0.792	0.796	0.803	0.811
$f_{\rm NL} = +2$	0.796	0.800	0.803	0.809	0.817
USR $\mu_* = 5/6$	0.808	0.816	0.825	0.851	0.914
USR $\mu_* = 5/2$	0.779	0.785	0.789	0.797	0.809

Table 2. The amplitude ratio $R = A_*^{\rm peaks}/A_*^{\rm threshold}$ between peaks of the non-Gaussian compaction, eq. (2.8), and threshold statistics, for lognormal spectra of width Δ, $f_{\rm PBH} = 1$ and $k_* = 10^7\,{\rm Mpc}^{-1}$, in the conventions of table 1.

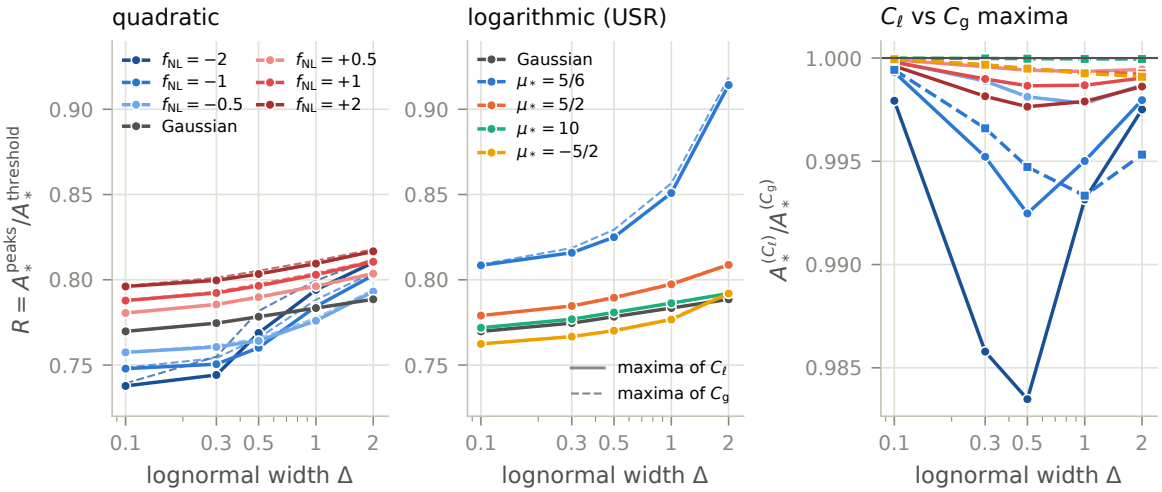


Figure 2. The amplitude ratio $R = A_*^{\rm peaks}/A_*^{\rm threshold}$ for $f_{\rm PBH} = 1$ against the lognormal width Δ, for the quadratic (left) and logarithmic (middle) templates. Solid lines count maxima of the non-Gaussian compaction C_ℓ, eq. (2.8); dashed lines count maxima of its Gaussian component $C_{\rm g}$, eq. (2.6). Right: the ratio $A_*^{(C_\ell)}/A_*^{(C_{\rm g})}$ of the amplitudes from the two kinds of maxima, with the colours of the left panels; the logarithmic family is dashed with square markers. Conventions of table 1, $k_* = 10^7\,{\rm Mpc}^{-1}$.

and $f_{\rm NL} = -2$ to -0.5, lognormal $\Delta = 0.1$–2). For the curvaton with $r_{\rm dec} = 0.9$, R moves by up to 6%. Its PBHs come from both branches of the map, with $R = 0.75$–0.79 from peaks of $\zeta_{\rm g}$ and 0.81–0.84 from troughs. On each branch alone R moves by at most 0.6%, but the threshold shifts the balance between them: at $\Delta = 0.3$ the troughs give 3%, 39% and 92% of $f_{\rm PBH}$ in threshold statistics for $C_{\rm th} \times 0.95$, 1 and 1.05.

Earlier comparisons for Gaussian ζ used other conventions. In their own conventions, BBKS peaks need 9–15% smaller amplitudes than Press–Schechter in ref. [36] and 11–12% in the `pbhstat` paper [29], and the high-peak formula of ref. [13] gives $R \approx 0.6$–0.67. Our $R_{\rm Gauss} = 0.77$–0.79 lies within the spread of the normalisation conventions above. The non-linear statistics of the compaction function [26, 54] go the other way: in `pbhstat` they need 4–12% larger amplitudes than Press–Schechter, mostly because of their profile-dependent threshold [29]. The prescription therefore sets the sign of the gap, not only its size.

5 Consequences for the PTA tension

5.1 Gravitational waves and the NANOGrav 15 likelihood

For Gaussian ζ in radiation domination the induced GWs are [7, 12, 55]

$$\Omega_{\rm GW,RD}(k) = \frac{1}{6}\int_1^\infty \mathrm{d}t \int_0^1 \mathrm{d}s \left[\frac{(t^2-1)(1-s^2)}{t^2-s^2}\right]^2 I^2_{t,s}\, \mathcal{P}_\zeta\Big(k\frac{t+s}{2}\Big)\mathcal{P}_\zeta\Big(k\frac{t-s}{2}\Big), \tag{5.1}$$

$$h^2\Omega_{\rm GW}(f) = h^2\Omega_r\, \frac{g_*}{g_*^0}\Big(\frac{g_{*s}}{g_{*s}^0}\Big)^{-4/3}\Omega_{\rm GW,RD}(k), \tag{5.2}$$

with $f = k/2\pi$, $h^2\Omega_r = 4.2\times10^{-5}$ and the Standard-Model $g_*(T)$ of ref. [56]. The kernel is [7, 12]

$$\begin{aligned} I^2_{t,s} = \frac{288(s^2+t^2-6)^2}{(t^2-s^2)^6}\Bigg[&\frac{\pi^2}{4}(s^2+t^2-6)^2\Theta(t-\sqrt{3}) \\ &+\Big(t^2-s^2-\tfrac{1}{2}(s^2+t^2-6)\ln\Big|\frac{t^2-3}{3-s^2}\Big|\Big)^2\Bigg]. \end{aligned} \tag{5.3}$$

We rewrite the logarithmic bracket as $(3-s^2)\,g(e)$ with $e = (t^2-s^2)/(3-s^2)$ and $g = -e^3/12 - e^4/12 - \ldots$, so that the cancellation at $t, s \to 1$ is done analytically, and we split t at the resonance $t = \sqrt{3}$. A flat spectrum gives $0.8222436\,A^2$ [7], the Dirac closed form is recovered, direct two-dimensional integration agrees to 10^{-8}–10^{-12}, and the infrared k^3 tail matches its asymptotic form. This is the spectrum for Gaussian ζ, as in ref. [12]; section 5.4 adds the primordial-NG corrections.

We use the NANOGrav 15-year free spectrum with Hellings–Downs correlations (the lowest 14 bins), represented by kernel density estimates (KDEs) of the per-bin posteriors in $\log_{10}\rho$ [57, 58]:

$$\ln L = \sum_j \ln p_j(\log_{10}\rho_j), \qquad \rho_j^2 = \frac{H_{100}^2\, h^2\Omega_{\rm GW}(f_j)}{8\pi^4 f_j^5 T}, \qquad T = 16.03\,\mathrm{yr}. \tag{5.4}$$

The KDEs use a compact kernel with bandwidths down to 0.004 dex. Outside the sampled range their density is floored, and their sparse tails are combs of single-sample spikes. We therefore regularise them by taking the maximum of each density and its 0.05-dex Gaussian blur (appendix D). This leaves the bulk unchanged and moves every significance below by $\leq 0.14\sigma$. The likelihood reproduces published fits. For $h_c \propto f^{-2/3}$ it gives $A = 2.52^{+0.59}_{-0.61}\times10^{-15}$ (90%), against $2.4^{+0.7}_{-0.6}\times10^{-15}$ from NANOGrav [1], and with a free slope $\gamma = 3.07^{+0.63}_{-0.56}$, against 3.2 ± 0.6. For the lognormal SIGW fit with Δ marginalised it gives $\log_{10}A = -0.57^{+0.54}_{-0.84}$, against $-0.56^{+0.53}_{-0.74}$ in ref. [12], and for the broken power law with the shape marginalised $\log_{10}A = -0.84^{+0.79}_{-0.74}$, against $-0.86^{+0.79}_{-0.65}$, and $\log_{10}k_* = 7.95$, against 7.92.

The posterior is taken in $(\log_{10}k_*, \log_{10}A)$, uniform on $[5,10]\times[-3,0]$, for the same spectral shape as the PBH line. The "allowed region" is $A < A_*(k_*)$, where $f_{\rm PBH}(A_*) = 1$. We quote two measures of its exclusion. The first, σ_P, is the one-sided Gaussian equivalent of the posterior mass in the allowed region. The second, $\sigma_{\rm HPD}$, is the two-sided equivalent of the highest-posterior-density level at which the region is first reached; this is the definition of ref. [12], whose $1\sigma/2\sigma/3\sigma$ contours lie at our 69/94/99% levels. Since $\sigma_{\rm HPD}$ saturates on local features of a lumpy posterior ridge, we lead with σ_P. Values above 4σ lie where the KDEs carry no information and are quoted only as ">4". The lines $A_*(k_*)$ for all k_* come from one set of β grids, because the moments depend on kr only.

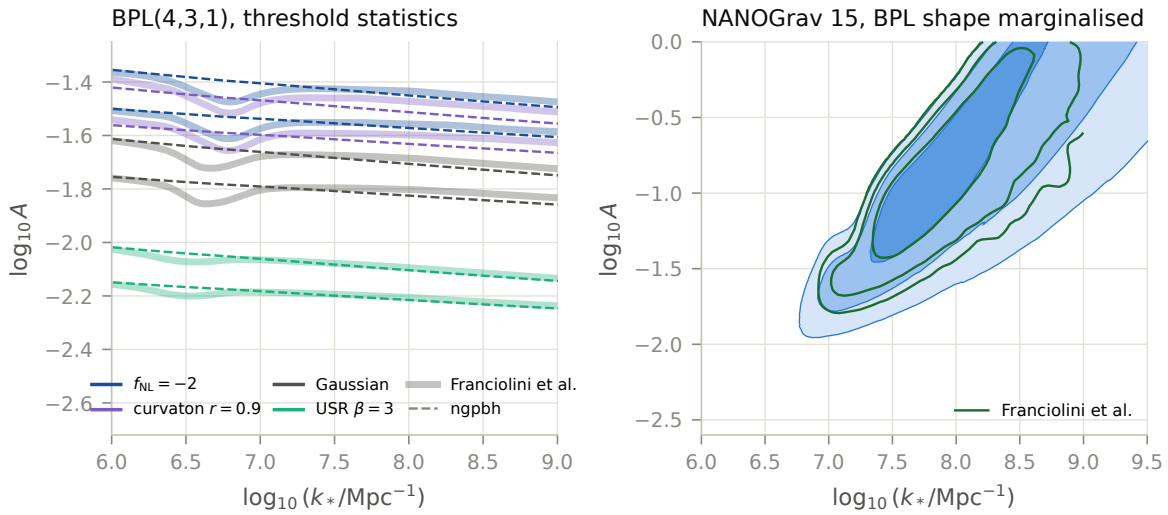


Figure 3. Comparison with ref. [12]. Left: threshold-statistics lines $f_{\rm PBH} = 1$ (upper of each pair) and 10^{-3} (lower) in the plane of the peak scale k_* and amplitude A of BPL(4,3,1), for four NG templates; wide lines are figure 3 of ref. [12] (with QCD effects), dashed lines are `ngpbh` (without them), in the conventions of ref. [12]. Right: the NANOGrav 15 posterior with the BPL shape marginalised (fill: 68.3/95.4/99.7% highest-posterior-density regions) and the contours of ref. [12].

5.2 Reproducing the bands of Franciolini et al.

Figure 3 compares our threshold-statistics bands with figure 3 of ref. [12], read exactly from the vector paths of the arXiv source. Their bands are reproduced only with two of their conventions: the superhorizon threshold of Musco et al., which their figure S3 confirms (appendix C), and their $f_{\rm PBH}$ normalisation. With both, the agreement for $k_* \geq 10^{7.5}\,\mathrm{Mpc}^{-1}$ is 1–6% in A for the Gaussian case (non-linearities only), within 6% for $f_{\rm NL} = -2$, within 2.3% for USR with $\beta = 3$, and 8–11% for the curvaton with $r_{\rm dec} = 0.9$. Their bands dip by up to 0.05 dex near $k_* \simeq 10^{6.5}\,\mathrm{Mpc}^{-1}$, where they include the QCD softening of the formation parameters and we do not. Putting their own lines on our posterior changes the significance by $\leq 0.07\sigma$, so the QCD omission does not matter, as they state.

5.3 Exclusion per statistic and NG template

Table 3 gives σ_P for threshold statistics and for peaks of the non-Gaussian compaction in the conventions of ref. [12], and figure 5 shows all templates. Over the 81 cases with $\sigma_P \leq 4$, peaks theory adds $\Delta\sigma_P = +0.24$ to $+0.78$ (median $+0.42$). The amplitude shift itself is nearly uniform, -0.07 to -0.11 dex (-0.05 for the strongest USR tail at $\Delta = 2$), and it translates into different $\Delta\sigma$ depending on where the line meets the posterior (figure 4). In the setup of ref. [12], with the shape-marginalised posterior and the BPL(4,3,1) lines, the Gaussian case goes from 2.3 to $2.8\sigma_P$ (1.4 to $1.9\sigma_{\rm HPD}$), $f_{\rm NL} = -2$ from 1.6 to $1.9\sigma_P$ and the curvaton from 1.7 to $2.0\sigma_P$. The USR case with $\beta = 3$, which they place beyond 3σ, gives $3.9\sigma_P$ with threshold statistics and > 4 with peaks. True NG peaks and the Gaussian-peak approximation agree to $|\Delta\sigma_P| \leq 0.04$.

The effect of NG itself is much larger. Relative to the Gaussian case it runs from -1.2σ (negative $f_{\rm NL}$ and the curvaton, for narrow spectra) to well beyond $+2\sigma$ (positive $f_{\rm NL}$ and USR tails). For $\Delta = 2$ the sign of the $f_{\rm NL} < 0$ shift flips, because large negative $\zeta_{\rm g}$ also produce large compaction.

template	BPL(4,3,1)	LN $\Delta = 0.5$	LN $\Delta = 1$	LN $\Delta = 2$
Gaussian	$2.98 \to 3.62$	$3.02 \to 3.52$	$3.54 \to > 4$	$3.84 \to > 4$
$f_{\rm NL} = -2$	$1.87 \to 2.18$	$1.88 \to 2.31$	$3.43 \to 3.84$	> 4
$f_{\rm NL} = -1$	$1.98 \to 2.40$	$1.90 \to 2.36$	$3.05 \to 3.46$	$3.89 \to > 4$
$f_{\rm NL} = -0.5$	$2.25 \to 2.93$	$2.22 \to 2.89$	$3.09 \to 3.51$	$3.60 \to > 4$
$f_{\rm NL} = +0.5$	$3.67 \to > 4$	$3.60 \to > 4$	> 4	> 4
curvaton $r_{\rm dec} = 0.9$	$2.01 \to 2.44$	$2.00 \to 2.43$	$3.09 \to 3.49$	$3.91 \to > 4$
USR $\mu_* = 5/2$	$3.53 \to > 4$	$3.49 \to 3.91$	> 4	> 4
$f_{\rm NL} \geq +1$, USR $\mu_* \leq 5/6$	> 4	> 4	> 4	> 4

Table 3. Exclusion of the PBH-allowed region by NANOGrav 15: σ_P, the one-sided Gaussian equivalent of the posterior mass in the region where $f_{\rm PBH} < 1$, for threshold statistics $\to$ peaks of C_ℓ, eq. (2.8). Conventions of ref. [12] (threshold without the transfer function, their $f_{\rm PBH}$ normalisation), with the same fixed spectral shape in the line and in the posterior (LN: lognormal).

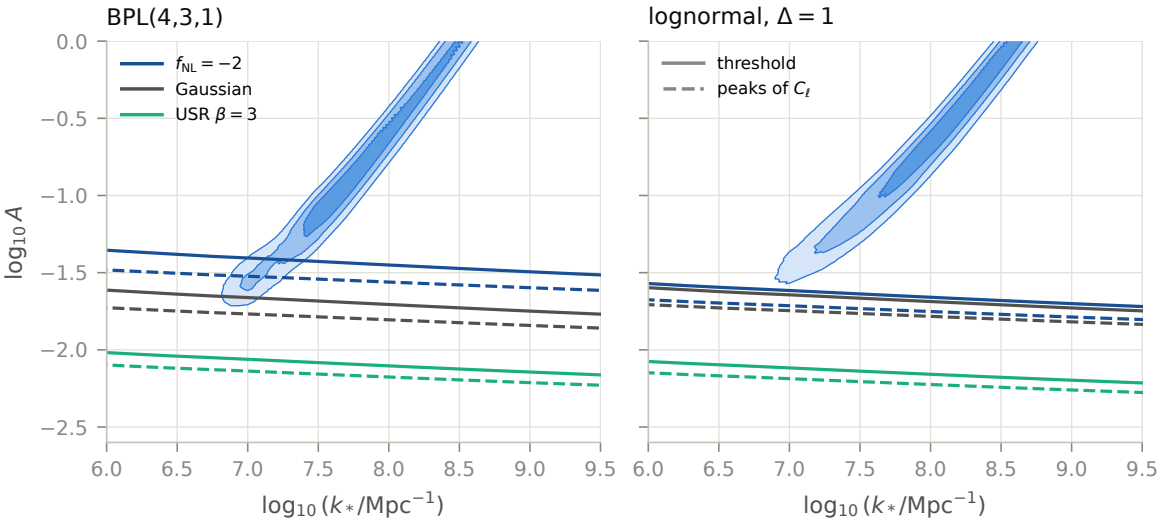


Figure 4. NANOGrav 15 posteriors in the plane of the peak scale k_* and amplitude A (68.3/95.4/99.7% highest-posterior-density regions) for BPL(4,3,1) (left) and a lognormal of width $\Delta = 1$ (right), with the $f_{\rm PBH} = 1$ lines from threshold statistics (solid) and peaks of C_ℓ (dashed) for $f_{\rm NL} = -2$, Gaussian ζ and USR with $\beta = 3$, in the conventions of ref. [12]. PBHs are overproduced above each line. Table 3 gives the exclusion σ_P of these lines; USR with $\beta = 3$ is beyond 4σ for both statistics.

5.4 Primordial NG in the induced GWs

So far the GW spectrum is that of Gaussian ζ, as in ref. [12]. For the quadratic template $\zeta = \zeta_{\rm g} + F(\zeta_{\rm g}^2 - \langle\zeta_{\rm g}^2\rangle)$ with $F = 3f_{\rm NL}/5$, the four-point function of ζ ends at F^4, and the spectrum has seven terms [41] (earlier computations include refs. [39, 40, 59]):

$$\Omega_{\rm GW,RD} = A^2 T_2 + F^2 A^3 \left(T_{\rm hyb} + T_C + T_Z\right) + F^4 A^4 \left(T_{\rm red} + T_{\rm pl} + T_{\rm np}\right). \tag{5.5}$$

For $\mathcal{P}_\zeta^{\rm g} = A\, s(k/k_*)$ each term depends on k/k_* only, and A is the amplitude of the Gaussian component, as in the PBH lines. All seven follow from

$$\Omega_X = \frac{k^3 c_X}{12\pi^2} \int \prod_{\rm loops} \frac{{\rm d}^3 q}{(2\pi)^3}\, {\rm Re}\left[z_1^2 \bar{z}_2^2\right] J_{12} \prod_i P(|\boldsymbol{m}_i|), \qquad J_{12} = \tfrac{1}{2}\left(I_{A1} I_{A2} + I_{B1} I_{B2}\right). \tag{5.6}$$

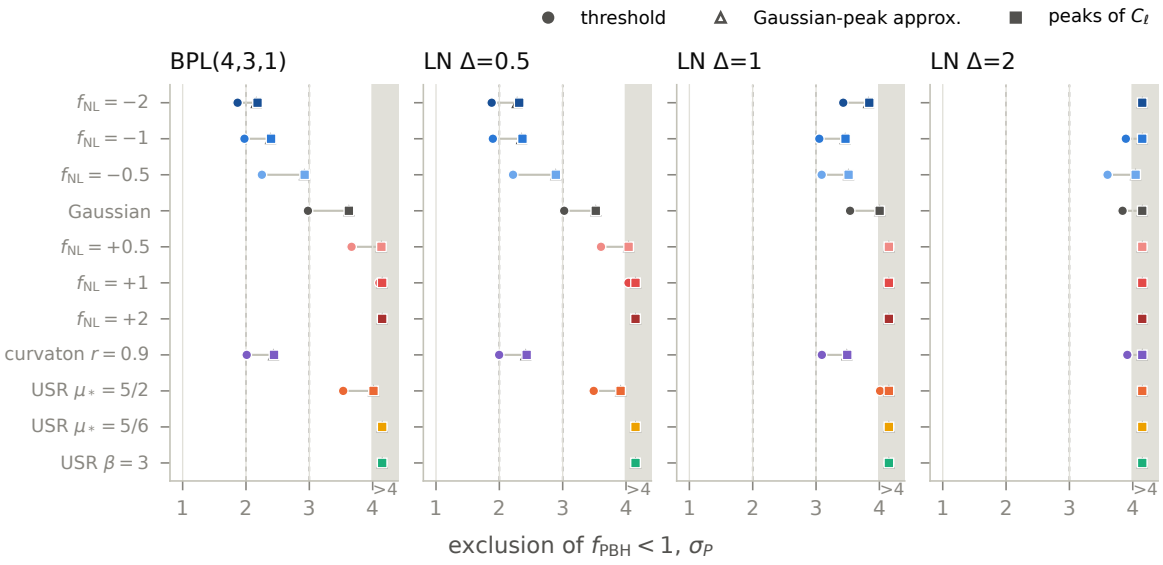


Figure 5. Exclusion σ_P of the PBH-allowed region by NANOGrav 15 for each NG template and spectral shape, with threshold statistics (circles), the Gaussian-peak approximation (open triangles, hidden behind the squares) and peaks of C_ℓ (squares), in the conventions of ref. [12] (LN: lognormal). Shaded: beyond 4σ, where the KDEs carry no information.

Here $z = (q_x + iq_y)/k$ is the transverse momentum of a GW vertex ($\boldsymbol{k} = k\hat{z}$). J_{12} is the oscillation average of the product of the late-time transfer functions $I = (I_A \sin x + I_B \cos x)/x$ of the two vertices [7], and J_{11} is the kernel above. The $\boldsymbol{m}_i$ are the momenta that carry a power spectrum, $P(m) = 2\pi^2 \mathcal{P}_\zeta^{\rm g}(m)/m^3$, and $c_X = 2$, 8, 16, 16, 8, 32, 16 for the Gaussian, hybrid, C, Z, reducible, planar and nonplanar terms. These reproduce the prefactors 2^5, 2^7F^2, 2^8F^2, 2^8F^2, 2^7F^4, 2^9F^4, 2^8F^4 of ref. [41].

The disconnected terms are the Gaussian functional of $P_{\rm g} + P_2$, with the one-loop spectrum $\mathcal{P}_\zeta^{(2)} = F^2A^2s_2$: we compute $T_{\rm hyb} = 2\,\Omega_{\rm bil}[s, s_2]$ and $T_{\rm red} = \Omega_{\rm G}[s_2]$ with the quadrature above. The connected terms are five- and eight-dimensional, and we evaluate them with randomised quasi-Monte Carlo over the spectral momenta. L of the $L+1$ momenta $\boldsymbol{m}_i$ are drawn from the spectrum, and the last follows from momentum conservation. The choices of which momentum is fixed are combined with the balance heuristic, so that the weights stay bounded when a momentum becomes soft. For broad spectra, pairs of nearly independent vertices dominate the variance: $\mathrm{Re}[z_1^2 \bar{z}_2^2] \propto \cos 2\phi$ averages to zero but scatters. We therefore average each sample over rotations about $\boldsymbol{k}$ of one spectral momentum (for the planar term, of two, independently). The rotations preserve the measure and every weight, so the estimate stays unbiased, and the oscillation cancels.

The Monte Carlo reproduces the quadrature of the Gaussian, hybrid and reducible terms in 72 comparisons with an rms pull of 0.97. All six NG terms agree with figure 4 of ref. [41], read from the vector paths of the arXiv source (figure 6): for lognormal widths 1/5 and 1/10 the agreement is within 3% of each curve's maximum, including the sign changes of T_Z and the negative $T_{\rm np}$.

For other templates, $\zeta = \zeta_{\rm g} + F\zeta_{\rm g}^2 + G\zeta_{\rm g}^3 + \ldots$, only F and G enter at $O(A^3)$. G enters through the power renormalisation $P \to (1 + 6G\sigma^2)P$, i.e. $+12G\sigma^2A^2T_2$; the $g_{\rm NL}$-type trispectrum vanishes under the azimuthal integral [60, 61]. $O(A^4)$ would need G^2, F^2G and

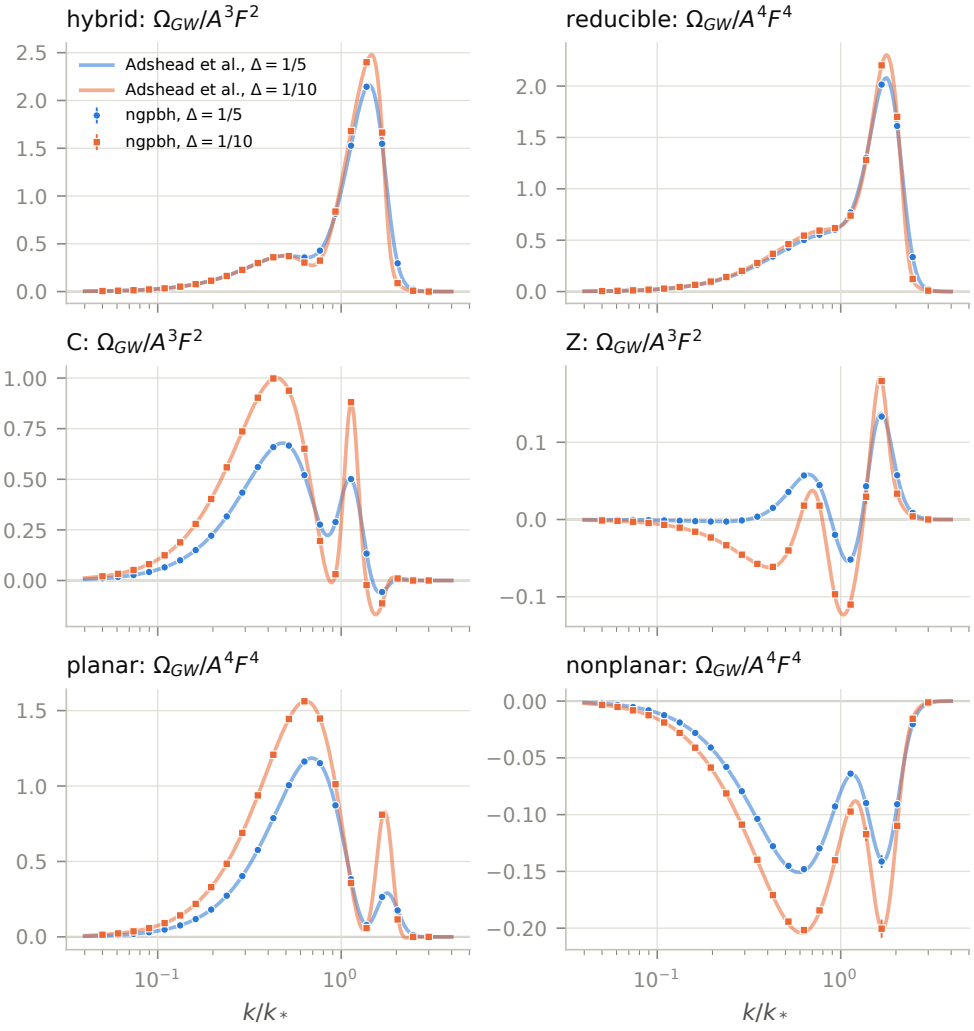


Figure 6. The six NG terms of the induced GW spectrum, eq. (5.5), for lognormal peaks of width $\Delta = 1/5$ and $1/10$ with $A = F = 1$: `ngpbh` (markers; the Monte Carlo errors are smaller than the markers) against figure 4 of Adshead, Lozanov and Weiner [41] (lines, read from the vector paths of the arXiv source).

FH terms, so we truncate the curvaton ($F = -0.62$, $G = 0.32$ [44]) and the USR maps ($F = 1/2\mu_*$, $G = 1/3\mu_*^2$) at A^3. The quadratic template is exact at A^4.

Relative to T_2 the corrections are large (figure 7). In the NANOGrav band of the posterior bulk, $k/k_* \simeq 0.02$–0.3, the ratio T_3/T_2 is 2–6. It grows with the width of the peak, because the one-loop spectrum collects the whole variance $\sigma^2 = A$ while the peak value of a broad lognormal is low. For $|f_{\rm NL}| = 2$, $\Omega_{\rm GW}$ at fixed A is therefore 10–40% larger near the PBH lines ($A \simeq 0.03$–0.05), and in the posterior bulk it is more than doubled. The posterior median of $\log_{10} A$ moves down by 0.08–0.18 dex for $|f_{\rm NL}| = 2$, by 0.04–0.11 for the curvaton, and by ≤ 0.02 for $|f_{\rm NL}| = 0.5$.

The PBH-allowed region lies at small A, so the exclusion weakens (table 4, figure 8). Over the cases with $\sigma_P \leq 4$, σ_P drops by 0.10–0.26 for $f_{\rm NL} = \pm 2$, by 0.03–0.20 for the curvaton, by ≤ 0.09 for $|f_{\rm NL}| = 1$, and by ≤ 0.03 for $|f_{\rm NL}| = 0.5$ and USR $\mu_* = 5/2$.

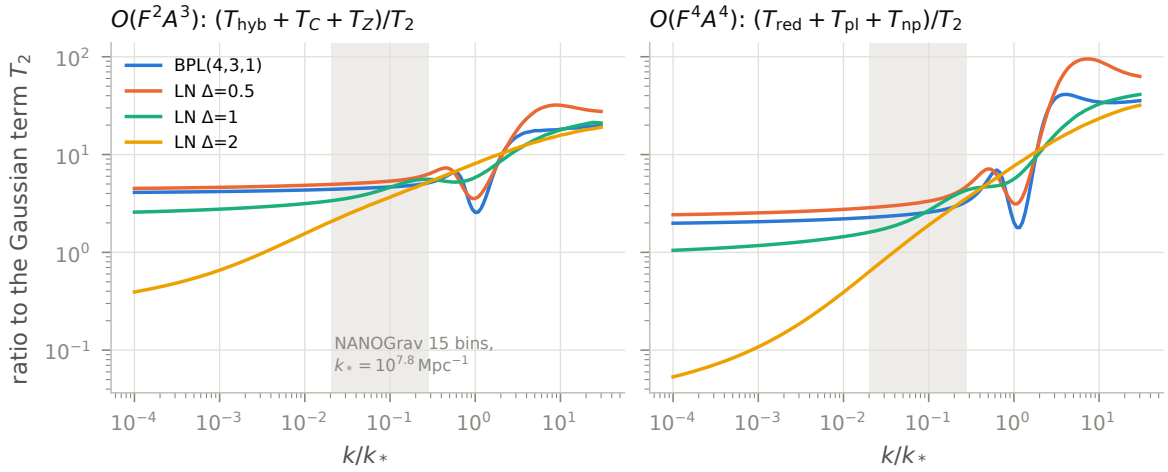


Figure 7. The NG corrections to the induced GW spectrum relative to the Gaussian term T_2, for each spectral shape: the $O(F^2A^3)$ terms (left) and the $O(F^4A^4)$ terms (right) of eq. (5.5), for the quadratic template with $F = 3f_{\rm NL}/5$. Shaded: the 14 NANOGrav bins for $k_* = 10^{7.8}\,{\rm Mpc}^{-1}$, near the posterior median.

template	GW spectrum	BPL(4,3,1)	LN $\Delta = 0.5$	LN $\Delta = 1$
$f_{\rm NL} = -2$	Gaussian	$1.87 \to 2.18$	$1.88 \to 2.31$	$3.43 \to 3.84$
	with NG	$1.73 \to 2.05$	$1.73 \to 2.11$	$3.25 \to 3.67$
$f_{\rm NL} = -1$	Gaussian	$1.98 \to 2.40$	$1.90 \to 2.36$	$3.05 \to 3.46$
	with NG	$1.94 \to 2.34$	$1.85 \to 2.30$	$2.99 \to 3.42$
$f_{\rm NL} = -0.5$	Gaussian	$2.25 \to 2.93$	$2.22 \to 2.89$	$3.09 \to 3.51$
	with NG	$2.24 \to 2.91$	$2.20 \to 2.87$	$3.07 \to 3.50$
curvaton $r_{\rm dec} = 0.9$	Gaussian	$2.01 \to 2.44$	$2.00 \to 2.43$	$3.09 \to 3.49$
	with NG	$1.92 \to 2.30$	$1.88 \to 2.26$	$2.95 \to 3.38$
USR $\mu_* = 5/2$	Gaussian	$3.53 \to > 4$	$3.49 \to 3.91$	> 4
	with NG	$3.51 \to 4.00$	$3.47 \to 3.90$	$3.99 \to > 4$

Table 4. Effect of primordial NG in the induced GWs on the exclusion: σ_P, the one-sided Gaussian equivalent of the posterior mass in the PBH-allowed region, for threshold statistics $\to$ peaks of C_ℓ, with the Gaussian induced GW spectrum (as in table 3) and with its primordial-NG corrections (each template with its own posterior). $f_{\rm PBH} = 1$, conventions of ref. [12]. The other templates are beyond 4σ with either spectrum.

The statistics shift is unchanged: threshold $\to$ peaks adds $+0.23$ to $+0.67$ (median $+0.39$), against $+0.24$ to $+0.68$ (median $+0.42$) with the Gaussian spectrum, and none of the 66 pairs changes by more than ± 0.10. The truncation matters little: dropping the A^4 terms of the quadratic template, or adding the F^4A^4 terms to the others, moves σ_P by ≤ 0.02. The GW part of the NG effect thus relieves the tension for either sign of $f_{\rm NL}$, by at most 0.26σ, while the PBH part shifts σ_P by -1.2 to beyond $+2\sigma$.

5.5 Robustness

Table 5 varies one analysis choice at a time. The statistics shift survives every change: it is $+0.2$ to $+0.8\sigma$ across the peak-normalisation conventions in use, and $+0.8$ to $+1.3\sigma$ with the high-peak formula. The absolute significance depends mainly on the frequency range,

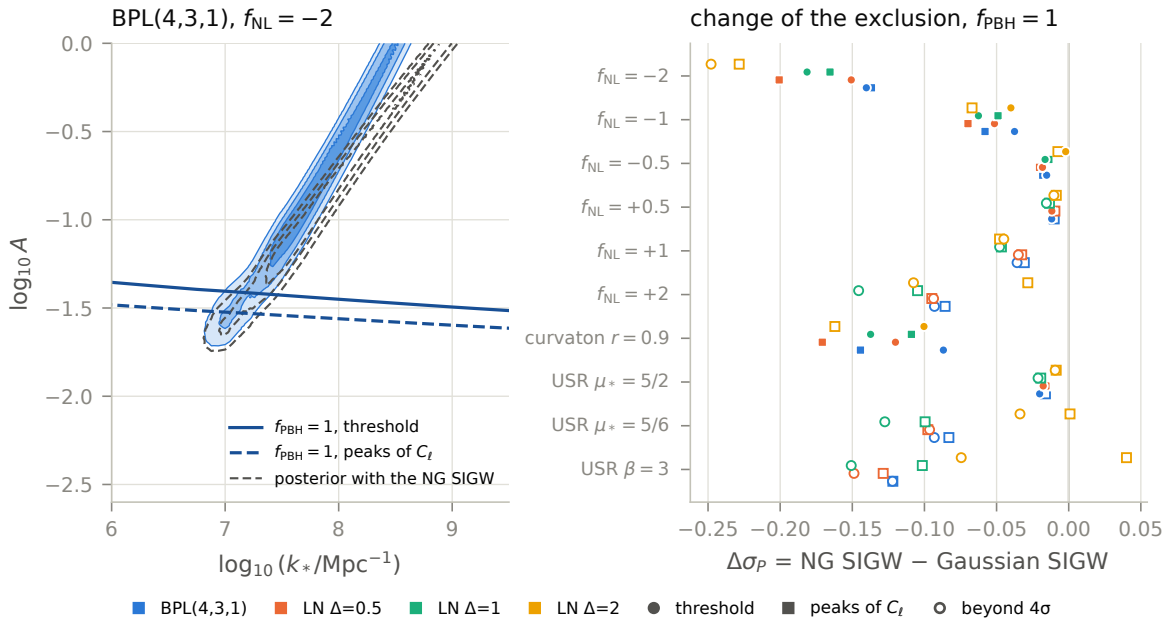


Figure 8. Effect of primordial NG in the induced GWs. Left: the NANOGrav 15 posterior for BPL(4,3,1) with the Gaussian induced GW spectrum (fill) and with its NG corrections for $f_{\rm NL} = -2$ (dashed; 68.3/95.4/99.7% highest-posterior-density regions), with the $f_{\rm PBH} = 1$ lines. Right: the change of σ_P from the NG corrections for every template (rows), spectral shape (colours) and statistic (circles: threshold statistics, squares: peaks of C_ℓ); open symbols lie beyond 4σ. Conventions of ref. [12], $f_{\rm PBH} = 1$.

as 5, 14 or 30 bins move it by -1.5 to $+1\sigma$: the part of the posterior nearest the allowed region, $k_* \simeq 10^7\,{\rm Mpc}^{-1}$, places the GW peak at 10–20 nHz, and the bins above it bound it. Among the other conventions, the horizon-crossing transfer function in the threshold moves the significance by -0.2 to -1.0σ, the entropy-conserving $f_{\rm PBH}$ normalisation by -0.15 to -0.4σ, $f_{\rm PBH} = 10^{-3}$ instead of 1 by $+0.4$ to $+0.8\sigma$, and a $\pm5\%$ change of $C_{\rm th}$ in both statistics by up to $\pm0.54\sigma$ for the Gaussian, $f_{\rm NL} < 0$ and curvaton templates. Fixing the spectral shape raises the significance relative to marginalising over it (Gaussian BPL: 3.0 against $2.3\sigma_P$), because the fixed BPL(4,3,1) has a steep causality tail that disfavours the low-k_* corner closest to the allowed region.

5.6 An astrophysical background

So far the SIGW is the whole signal. We now add a power law from supermassive black hole binaries (SMBHBs), whose timing residuals add to those of the SIGW, $\rho_j^2 \to \rho_j^2 + A_B^2 (f_j/f_{\rm yr})^{3-\gamma_B}/(12\pi^2 f_j^3 T)$, with NANOGrav's astrophysical prior on $(\log_{10} A_B, \gamma_B)$, a two-dimensional Gaussian with mean $(-15.6, 4.7)$ [9], and marginalise over it. SIGW plus SMBHB is preferred over the SMBHB alone by Bayes factors of 94–105 for BPL(4,3,1) and the log-normals with $\Delta \leq 1$, and 38 for $\Delta = 2$, of the same order as NANOGrav's 23–59 for their SIGW templates [9]. These Bayes factors, like NANOGrav's, rest on log-uniform priors for the intrinsic red noise of the pulsars, which inflate the amplitude of the background through a prior-volume effect: with priors uniform in power or informed by the pulsar population, the preference for scalar-induced waves over SMBHBs falls to order unity [62].

change from the reference	$\Delta\sigma_P$, threshold $\to$ peaks	max. change of σ_P
none (14 bins, KDE blur 0.05 dex)	+0.31 to +0.68	—
prior $\log_{10} k_* \in [6, 9]$, or $\log_{10} A \geq -4$	unchanged	0.00
posterior grid twice as fine	unchanged	0.00
KDE blur 0, 0.03 or 0.10 dex	+0.30 to +0.72	0.14
5 frequency bins (2–10 nHz)	+0.27 to +0.72	1.45
30 frequency bins (2–59 nHz)	+0.23 to +0.55	1.06
peak volume $(2\pi)^{3/2} r^3$	+0.40 to +0.81	0.14
Press–Schechter factor 2	+0.27 to +0.59	0.09
`pbhstat` v0.0.6 normalisation	+0.16 to +0.39	0.33
high-peak formula [13] (indicative)	+0.82 to +1.29	0.65
$C_{\rm th} \times 0.95$ (five templates)	+0.42 to +0.75	0.54
$C_{\rm th} \times 1.05$ (five templates)	+0.23 to +0.72	0.54

Table 5. Robustness of the PTA significances to one analysis choice at a time, for BPL(4,3,1) and lognormals with $\Delta = 0.5$ and 1, over seven NG templates (cases with $\sigma_P \leq 4$); the $C_{\rm th}$ rows cover the Gaussian, $f_{\rm NL} = -2, -1, -0.5$ and curvaton templates. Reference: the conventions of ref. [12] and the uniform prior $\log_{10} k_* \in [5, 10]$, $\log_{10} A \in [-3, 0]$.

The PBH-allowed region now contains the solutions in which the SMBHB alone explains the signal. As a result σ_P is capped at 2.3–2.8σ for every template, or 2.0–2.5σ if the prior on $\log_{10} A$ extends to -4, and the templates that relieve the tension stay at 1.8–2.6σ. The statistics shift drops to +0.04 to +0.39σ (median +0.26, against +0.43 without the SMBHB on the same grid). The profile $\Delta\chi^2$, which does not depend on the prior range, keeps the shift where the PBH line cuts the SIGW posterior (7.7 to 12.5 for the Gaussian BPL) and saturates at $\simeq 18$, the cost of explaining the signal with the SMBHB alone.

6 Discussion

The question we set out to answer is whether the gap between threshold statistics and peaks theory, once local NG is included for broad spectra, changes how strongly PBH overproduction disfavours a scalar-induced explanation of the PTA signal. It does, modestly and uniformly. Peaks theory lowers the overproduction line by about 0.1 dex for every NG template, which strengthens the exclusion by $\simeq 0.4\sigma$, or 0.2–0.8σ depending on the peak normalisation. In the framing of ref. [13] the same effect relaxes the PTA bound on $f_{\rm PBH}$: at fixed A peaks theory predicts more PBHs, so the PTA bound on A allows a larger $f_{\rm PBH}$, while the line $f_{\rm PBH} = 1$ moves down into the PTA posterior. Their high-peak normalisation ($R \approx 0.6$–0.67) enlarges both effects.

The NG dependence of the gap cancels almost completely, the double ratio being 1 to about 4%, so NG and the choice of statistics add rather than interact. Negative $f_{\rm NL}$ and the curvaton remain the ways to relieve the tension, but peaks theory takes back 0.3–0.5σ of the relief. NG in the induced GWs adds power at fixed A; it relieves the tension by up to 0.26σ for $|f_{\rm NL}| \leq 2$ and does not change the gap between the statistics. None of the conclusions of ref. [12] changes qualitatively: their remark on peaks theory is correct, and now quantified.

For an absolute statement about the PTA tension, the larger uncertainties lie elsewhere. The largest is an astrophysical SMBHB background: with NANOGrav's prior it caps every exclusion at 2.3–2.8σ and reduces the statistics shift to 0.04–0.39σ. The priors on the red noise of the pulsars act in the same direction: those behind the free-spectrum densities we use

inflate the amplitude of the background [62], and priors that avoid this effect would move the posterior towards the PBH-allowed region. Among the analysis choices, the frequency range of the fit, the treatment of the collapse threshold and the $f_{\rm PBH}$ normalisation each move the significance by about as much as the choice of statistics, or more. All three are rarely stated, and we recommend quoting them with any PBH bound on the PTA signal, together with the peak normalisation when peaks theory is used. On the theory side, the Gaussian-peak approximation is adequate for the PTA question (to 0.04σ), although it miscounts peaks by up to 18%.

Our results rest on several simplifications. We assume spherical symmetry for the compaction of a peak [30] and use the mean profile, as the literature we compare with does; profile dispersion can matter for negative NG [42]. The collapse threshold follows from the power spectrum and is the same for every NG template, although NG also deforms the typical profile and hence the threshold [52, 53]; a $\pm5\%$ change moves σ_P by up to 0.54σ and R by $\leq 0.3\%$ ($\leq 6\%$ for the curvaton, section 4). Each scale is counted separately, without a cloud-in-cloud correction. We neglect QCD effects on the formation parameters [63, 64] and on the GW spectrum [65, 66], truncate the NG series of the GW spectrum at A^3 for the curvaton and USR maps, and use the free-spectrum refit, with NANOGrav's priors on the red noise of the pulsars, rather than the full PTA likelihood; the SMBHB background is a power law with NANOGrav's prior. These choices are common to both statistics, so the comparison between them is like-for-like, while the absolute numbers carry the systematics of table 5. Finally, R refers to BBKS maxima of the compaction at fixed scale, the prescription that differs from threshold statistics only by the peak condition. Peaks of ζ with typical profiles [24, 25, 31] and the non-linear statistics of ref. [26] define the abundance differently; we have not computed R for them in our conventions, but section 4 quotes what the literature finds for Gaussian ζ.

7 Conclusions

We have constructed peaks theory with local NG for broad spectra on the compaction function, as the exact counterpart of threshold statistics. Maxima of the non-Gaussian compaction are maxima of a shifted Gaussian field, corrected by a rank-one term, eq. (2.8), and Monte Carlo and lattice counts confirm this. Peaks require 18–26% smaller amplitudes than threshold statistics, and this ratio is almost independent of NG for $|f_{\rm NL}| \leq 2$: a threshold-statistics amplitude converts to peaks theory as $A_*^{\rm peaks} \simeq R_{\rm Gauss} A_*^{\rm threshold}$, with $R_{\rm Gauss} = 0.783 + 0.0066 \ln\Delta$ for the lognormal and 0.775–0.778 for the BPL, to about 4%.

For NANOGrav 15 this strengthens the exclusion of a scalar-induced explanation by 0.2–0.8σ for every NG template, or by 0.04–0.39σ with an SMBHB background, which caps the exclusion at 2.8σ. The effect is smaller than that of NG itself for $|f_{\rm NL}| \gtrsim 1$, and comparable to three analysis conventions, which we have quantified. The primordial-NG corrections to the induced GW spectrum, with all seven terms, relieve the tension by up to 0.26σ for $|f_{\rm NL}| \leq 2$ and leave the gap between the statistics unchanged. The code `ngpbh` implements every piece and reproduces published results: `pbhstat`, Ferrante et al., Franciolini et al. and Adshead et al.

A Peaks of the non-Gaussian compaction: the rank-one factor

At a stationary point of C_ℓ, write $H_\lambda = \nabla\nabla Y_\lambda$ and $\boldsymbol{g} = \nabla\zeta_r$, with $\boldsymbol{g} \sim \mathcal{N}(0, v\mathbb{I})$ independent of H_λ. The exact density is

$$n(c,z) = P_{\rm G}(c,z)\Big(\frac{3}{2\pi\sigma_{1,\lambda}^2}\Big)^{3/2}\mathbb{E}\big[|\det(H_\lambda + \kappa\boldsymbol{g}\boldsymbol{g}^T)|\,\Theta_s\big], \tag{A.1}$$

where Θ_s requires negative (positive) definiteness for $s = +1$ (-1). For $H \prec 0$ the matrix determinant lemma gives $\det(-H - \kappa\boldsymbol{g}\boldsymbol{g}^T) = \det(-H)\,[1 - \kappa\,\boldsymbol{g}^T(-H)^{-1}\boldsymbol{g}]$, which is linear in $\kappa|\boldsymbol{g}|^2$. Averaging over the direction and length of $\boldsymbol{g}$ gives $v\,\det(-H)\mathrm{tr}(-H)^{-1} = v\,e_2(-H)$. In the BBKS shape variables $e_2 = x^2/3 - (3y^2+z^2)/3$, so $f_2(x) = \mathbb{E}[e_2\Theta|x]$ follows from the same two-dimensional integral as f; that integral reproduces the BBKS function f to 10^{-10}. The interior part of the first-order factor is then exactly $1-\tau R_2$. For $\tau > 0$ maxima can only be lost, so $1-\tau R_2$ is a lower bound; for $\tau < 0$, configurations that become maxima add a positive $\mathcal{O}(\tau^2)$ term. The Monte Carlo bounds the remainder at 0.85% at the most extreme node tested, a shift of 0.02% in A_*.

B The PBH abundance from entropy conservation

After formation at temperature T, the PBH number per entropy is conserved, so $\rho_{\rm PBH,0} = \frac{3}{4}\beta\,(g_*/g_{*s})\,T\,s_0$, with $s_0 = \frac{2\pi^2}{45}g_{*s}^0 T_0^3$. Here $\beta(M_H)$ is the fraction of the energy density at formation that collapses into PBHs, weighted by $M/M_H = \mathcal{K}(C-C_{\rm th})^\gamma$ as in eq. (2.5). With $M_H = 4\pi M_P^2/H$ and $H = \pi(g_*/90)^{1/2}T^2/M_P$ this gives

$$f_{\rm PBH} = \frac{1}{\Omega_{\rm DM}}\int \mathrm{d}\ln M_H \Big(\frac{M_H}{M_\odot}\Big)^{-1/2}\Big(\frac{g_*}{106.75}\Big)^{3/4}\Big(\frac{g_{*s}}{106.75}\Big)^{-1}\frac{\beta(M_H)}{1.0\times10^{-8}}, \tag{B.1}$$

for $h = 0.674$ and $\Omega_{\rm DM}h^2 = 0.12$. It agrees with Carr et al. [67] to 2%, and with the exact expression of ref. [68] (their eq. II.3, written in terms of M_H) to 1.5%.

The constant 7.9×10^{-10} in eq. (14) of ref. [12] makes $f_{\rm PBH}$ 12.7 times larger. It equals $[2.4\,(M_{\rm eq}/M_\odot)^{1/2}]^{-1}$ with $M_{\rm eq} = 2.8\times10^{17}\,M_\odot$, the normalisation of the approximation $f_{\rm PBH} \simeq 2.4\,\beta\,(M_{\rm eq}/M)^{1/2}$ of ref. [68], which is already a fraction of the dark matter. Dividing it by $\Omega_{\rm DM}$ once more gives a factor 3.8, and the approximation itself exceeds the entropy-conserving result by 3.4 when M is identified with M_H. The form $(M_{\rm eq}/M_H)^{1/2}\beta/\Omega_{\rm DM}$ drops the factor Ω_m and the heating of the radiation by the changes of g_{*s} after formation, and is 3–5 times larger. Both lower the amplitude for $f_{\rm PBH} = 1$ by $\ln(\text{factor})/|\ln\beta|$, i.e. 8–12% at $|\ln\beta| \simeq 21$.

C The collapse threshold of Franciolini et al.

Figure S3 of ref. [12] plots $\delta_{\rm th}$ from the Musco et al. prescription against the lognormal width and against the UV slope β of BPL(4,β,1). The figure is vector graphics in the arXiv source, so we read the curves exactly and compare them with three readings of the prescription (figure 9). Without a transfer function we reproduce their curves to ≤ 0.0002 for $0.1 \le \Delta \le 1$ and to ≤ 0.0016 for $1.9 \le \beta \le 3$. With T at $\eta = \hat{r}_m/10$, the superhorizon smoothing of Musco et al., the agreement is ≤ 0.0005 and ≤ 0.0012. With T at horizon crossing, which is what the notation $T(k, r_m)$ of their eqs. (S4)–(S5) suggests, the curves are

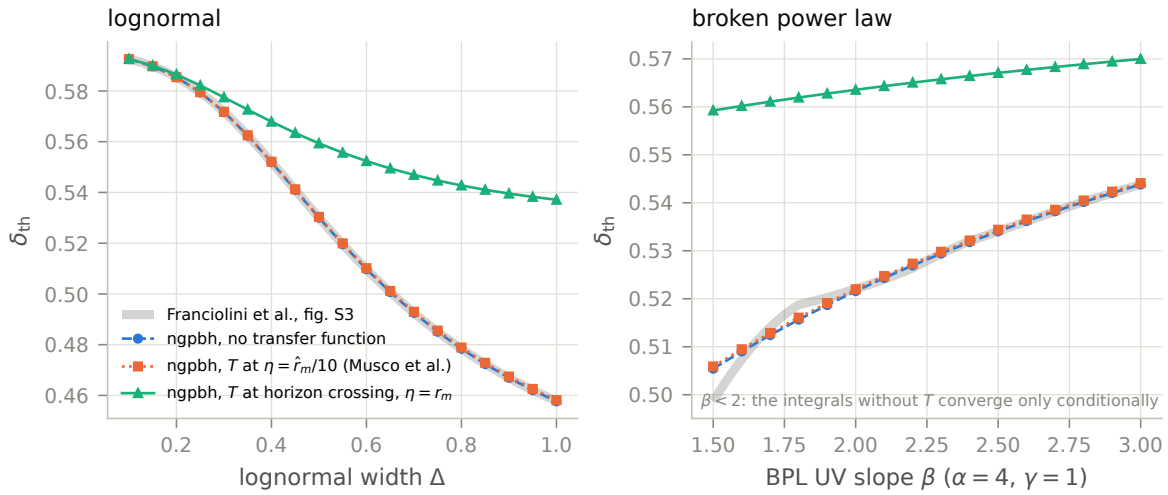


Figure 9. The collapse threshold $\delta_{\rm th}$ of figure S3 of ref. [12] (wide gray), read from its vector paths, against the lognormal width Δ (left) and the UV slope β of BPL(4,β,1) (right), compared with `ngpbh` for three readings of the Musco et al. prescription: without the transfer function, with Musco et al.'s superhorizon smoothing ($\eta = \hat{r}_m/10$) and with the horizon-crossing transfer function.

off by up to 0.08. For $\beta < 2$ the integrals without T converge only conditionally, because the UV tail falls like $k^{-\beta}$; their curve has a kink at $\beta \simeq 1.8$ and lies up to 0.007 below ours there. Their plotted α_s for the BPL is not the one behind their plotted $\delta_{\rm th}$ through their eq. (S6), while for the lognormal both curves agree with ours. Their thresholds are therefore those of Musco et al., and the conclusions of ref. [12] do not depend on the notation.

D Free-spectrum density estimates

The per-bin densities distributed with PTArcade [57, 58] are Epanechnikov KDEs on a grid of 10^4 points in $\log_{10}\rho$. The kernel has compact support, so the density is exactly zero (stored as $\ln p = -36$) outside the range of the samples, and within the sparse tails it alternates between single-sample spikes and zero. The two most constraining bins (3.9 and 5.9 nHz) have almost no low-ρ tail. Tail probabilities beyond $\sim 2.5\sigma$ are therefore lumpy and cliff-like. We replace each density by the maximum of itself and its convolution with a 0.05-dex Gaussian, which leaves the bulk unchanged and fills the holes. Table 5 shows that the result changes by at most 0.14σ between no blur and 0.10 dex.

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Code and data availability

The code `ngpbh` (MIT licence) is archived at Zenodo. Version 0.7.1, doi:10.5281/zenodo.23661308, contains the scripts that produce every number, table and figure of this paper

(`README.md`), 96 unit tests, the digitised figures 3 and S3 of ref. [12] and figure 4 of ref. [41], and the tables of the NG terms of the induced GW spectrum per spectral shape; doi:10.5281/zenodo.23624506 resolves to the latest version. The NANOGrav 15 free-spectrum densities are those of PTArcade (Zenodo record 10495907) and are downloaded with an MD5 check.

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